

Complete NWP Weather Feature Specification

Feature Group	Variable	Formula / Encoding	Dim	Dataset	Physical Meaning
Raw NWP (GEFCom2012)	U, V	—	2x5=10	GEFCom2012	Wind vector components
	Wind Speed	$\sqrt{U^2+V^2}$	1x5=5	GEFCom2012	Wind magnitude
	Wind Direction	atan2(U,V) [rad]	1x5=5	GEFCom2012	Wind bearing
Raw NWP (GEFCom2014)	U10, V10	—	2	GEFCom2014	10m wind vector (surface layer)
	U100, V100	—	2	GEFCom2014	100m wind vector (hub height)
Derived (GEFCom2014)	WS10, WS100	$\sqrt{U^2+V^2}$	2	GEFCom2014	Wind speed at 10m & 100m
	WD10, WD100	atan2(U,V)	2	GEFCom2014	Wind direction [radians]
	SHEAR	WS100/(WS10+0.1)	1	GEFCom2014	Atmospheric stability index
	VEER	sin(WD100-WD10)	1	GEFCom2014	Directional wind veer with height
Cyclic Time	HOUR_SIN/COS	sin/cos(2π*h/24)	2	Both	Diurnal cycle encoding
	MONTH_SIN/COS	sin/cos(2π*m/12)	2	Both	Seasonal cycle encoding

GEFCom2012: 5 lead times x 4 vars + 4 time = 24 dims. GEFCom2014: 10 weather + 4 time = 14 dims. Unified: 20+10+4 = 34 dims (zero-padded).